

Euclidean geometry – 2

Conics, areas and projective maps

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Theoretical minimum. A conic is seen metrically (focus and directrix), algebraically (a quadratic equation, classified by linear algebra) and projectively (a curve fixed by five points, carrying cross-ratios); for areas one first applies an affine map turning it into a circle, $xy = 1$ or $y = x^2$.

Definitions. For a point F , a line $l \not\ni F$ and $e > 0$ the *conic* $\{X : |XF| = e d(X, l)\}$ with focus F , directrix l and eccentricity e is an ellipse, a parabola or a hyperbola for $e < 1$, $e = 1$, $e > 1$; in polar coordinates at F , with φ measured from the perpendicular from F to l , it is $r = p/(1 + e \cos \varphi)$, $p = e d(F, l)$. The *signed area* is $[ABC] = \frac{1}{2} \det(B - A, C - A)$, positive for counterclockwise ABC . The *cross-ratio* of points a, b, c, d of a line is $(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}$; a *projective map* of $\mathbb{R} \cup \{\infty\}$ is $x \mapsto \frac{\alpha x + \beta}{\gamma x + \delta}$, $\alpha\delta \neq \beta\gamma$; a *perspectivity* from a line l to a line m with centre $O \notin l \cup m$ sends X to $OX \cap m$.

Theorem 1 (focal and reflective properties). For $e \neq 1$ there is a second focus and directrix, symmetric in the centre, and $|XF_1| + |XF_2| = 2a$ (ellipse), $||XF_1| - |XF_2|| = 2a$ (hyperbola), with $c = |OF_i| = ea$ and directrices at distance a/e from the centre O . The tangent at X bisects the external angle F_1XF_2 of an ellipse and the internal one of a hyperbola; for a parabola it bisects the angle between XF and the perpendicular from X to l . *Proof idea:* X minimises $|YF_1| + |YF_2|$ over the tangent, since the other points of the tangent lie outside; then use Heron's reflection. So light from one focus is reflected into the other one (parallel to the axis for a parabola).

Theorem 2 (equations, five points, pencils). A conic is the zero set of $Q = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f$; up to affine maps a non-degenerate non-empty real conic is $x^2 + y^2 = 1$, $x^2 - y^2 = 1$ or $y = x^2$ as $ac - b^2 > 0$, < 0 , $= 0$ (Linear algebra – 5), and projectively all three coincide. A line not contained in a conic meets it at most twice; as Q has six coefficients, through five points, no four collinear, passes exactly one conic. The conics through

four points, no three collinear, form a *pencil* $\lambda Q_1 + \mu Q_2$ containing exactly three line pairs: the two pairs of opposite sides and the pair of diagonals.

Theorem 3 (areas). In coordinates

$$[ABC] = \frac{1}{2}((x_B - x_A)(y_C - y_A) - (x_C - x_A)(y_B - y_A)),$$

so $P \mapsto [PBC]$ is affine and $[PBC] + [PCA] + [PAB] = [ABC]$ for all P (barycentric coordinates); at an interior extremum of $[PAB]$ over a smooth arc the tangent is parallel to AB . An affine map $x \mapsto Mx + v$ multiplies areas by $\det M$ and preserves area ratios, parallelism, tangency and centres; all parabolic segments are affinely equivalent, and $y = x^2$ cuts off over $[a, b]$ a segment of area $(b - a)^3/6$.

Theorem 4 (projective maps of a line). A perspectivity with centre O preserves the cross-ratio, since $(A, B; C, D) = \frac{\sin \angle AOC \sin \angle BOD}{\sin \angle BOC \sin \angle AOD}$ (oriented angles). Projective maps are exactly the cross-ratio preserving bijections of $\mathbb{R} \cup \{\infty\}$; each is determined by the images of three points, so one with three fixed points is the identity. Compositions of perspectivities are projective, and every projective map of a line to itself is a composition of at most three perspectivities (over $\mathbb{C}$: the Möbius maps of Complex analysis – 7).

Fact 1 (normals of a parabola, Apollonius). The normal to $x^2 = 2py$ at $(t, t^2/2p)$ passes through (u, v) iff $t^3 + 2p(p - v)t - 2p^2u = 0$: at most three normals pass through a point, the abscissas of their feet add up to 0, and there are three exactly when $8(v - p)^3 > 27pu^2$.

Fact 2 (billiards, Birkhoff). In a smooth strictly convex curve Γ an inscribed n -gon of maximal perimeter exists by compactness; at each vertex X the sum $|YX_-| + |YX_+|$ over $Y \in \Gamma$ is maximal at $Y = X$, so the normal bisects $\angle X_-XX_+$ and the polygon is a closed billiard trajectory — the argument of Theorem 1 again.

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1. Derive the polar equation from $|XF| = e d(X, l)$ and, for $e \neq 1$, the canonical equation $\frac{x^2}{a^2} \pm \frac{y^2}{b^2} = 1$ with $c = ea$ and the directrices $x = \pm a/e$; prove the focal-sum property of the ellipse and the reflective properties of Theorem 1, and show that a billiard ball sent through one focus of an elliptic table passes through the other focus after every reflection. Then prove Fact 1.

2. Prove the determinant formula of Theorem 3 and the identity $[PBC] + [PCA] + [PAB] = [ABC]$. Prove that a perspectivity preserves the cross-ratio, that a projective map of a line is determined by the images of three points, and deduce that a projective map other than the identity has at most two fixed points. Finally, prove that through five points of which no four are collinear there passes exactly one conic.
3. Two different ellipses are given. One focus of the first ellipse coincides with one focus of the second ellipse. Prove that the ellipses have at most two points in common. (IMC 2008)
4. The normals to the parabola $2y = ax^2$ ($a > 0$) at two different points B and B' of the parabola meet at a point A of the parabola, different from B and B' ; let m be the ordinate of A . Prove that $m > \frac{4}{a}$. (IMC 2025, shortlist)
5. An ellipse, a parabola and a hyperbola are circumscribed about a trapezoid. Prove that the line joining the centres of the ellipse and of the hyperbola is parallel to the axis of the parabola. (МФТИ 2016)